Interval search experiments

Built 2026-10-16 13:31 PT. Experiment design by Roland Gao; runs and analysis by Claude. Code: research/cooldown_search_probe.py (runs), research/cooldown_study.py (analysis), research/make_cooldown_study.py (this PDF). Logs: logs/research_cooldown_search_probe_*, logs/research_hparam_importance_*. Batch size 2000 throughout.

Experiment 1. Cooldown study: which 40-step lookahead ranks interval candidates like the final result?

Question

An interval search picks each interval's LR by training the interval with a candidate LR, then a short lookahead (the cooldown), and scoring the result. Which cooldown ranks the candidates the way the final accuracy does? Compared, all with a 40-step lookahead (one interval at batch size 2000): no lookahead; linear decay to 0 (conv only, or all four groups); one searched constant cooldown, either for conv only with the other groups on their ground-truth schedules, or shared by all four groups; and four searched constant cooldowns (one per group).

Setup

Item	Value
Model / data	CIFAR-10 airbench, batch size 2000, 200 steps, baseline_bs2000 (conv Muon2 0.22 -> 0; head, norm_bias, whiten_bias SGD on their baseline schedules)
Intervals	4 intervals of 40 steps: steps 0-40, 40-80, 80-120, 120-160 (the last interval has no room for a 40-step lookahead)
Candidates	7 constant conv LRs for the interval: the 0.6^k grid point nearest the baseline's mean conv LR in that interval, and 3 grid steps either side; other groups on the baseline
Ground truth	Final test TTA when training continues on the baseline schedule after the interval
No lookahead	Test TTA right after the interval
Linear to 0	Conv only: conv decays linearly from the candidate to 0 over 40 steps, the other groups continue their schedules. All groups: every group decays linearly to 0 from its current value
1 searched cooldown: conv only	Conv constant at m x candidate for 40 steps; the other groups follow their baseline (ground-truth) schedules; m chosen from {1, 0.6, 0.36, 0.22, 0.13, 0.078, 0.047, 0} by the best score on that seed
1 searched cooldown: shared	Constant LRs for 40 steps: conv at m x candidate, each other group at m x its mean LR over the interval; one m for all groups, chosen as above
4 searched cooldowns	As above but one m per group; coordinate search from the best shared m, one group at a time over the same 8 values, until no group changes (at most 3 passes)
Seed	1 training seed (0); every method uses the same prefix and data order
Score	Test-set TTA accuracy (%)

Results

Picked LR = the candidate with the highest score. Regret = ground truth at its best candidate minus ground truth at the picked one (pp). One seed: single-run noise is about 0.13 pp, so differences of that size between candidates are not resolved.

Picked LR (regret, pp)

Method	Interval 1	Interval 2	Interval 3	Interval 4
Ground truth (final TTA)	0.078 (0.00)	0.13 (0.00)	0.078 (0.00)	0.078 (0.00)
No lookahead	0.078 (0.00)	0.028 (0.34)	0.028 (0.01)	0.028 (0.29)
Linear to 0: conv only	0.13 (0.05)	0.078 (0.26)	0.078 (0.00)	0.028 (0.29)
Linear to 0: all four groups	0.13 (0.05)	0.078 (0.26)	0.047 (0.21)	0.028 (0.29)
1 searched cooldown: conv only, others on ground truth	0.13 (0.05)	0.078 (0.26)	0.047 (0.21)	0.047 (0.21)
1 searched cooldown: shared by all four groups	0.13 (0.05)	0.078 (0.26)	0.047 (0.21)	0.047 (0.21)
4 searched cooldowns (one per group)	0.13 (0.05)	0.078 (0.26)	0.047 (0.21)	0.047 (0.21)

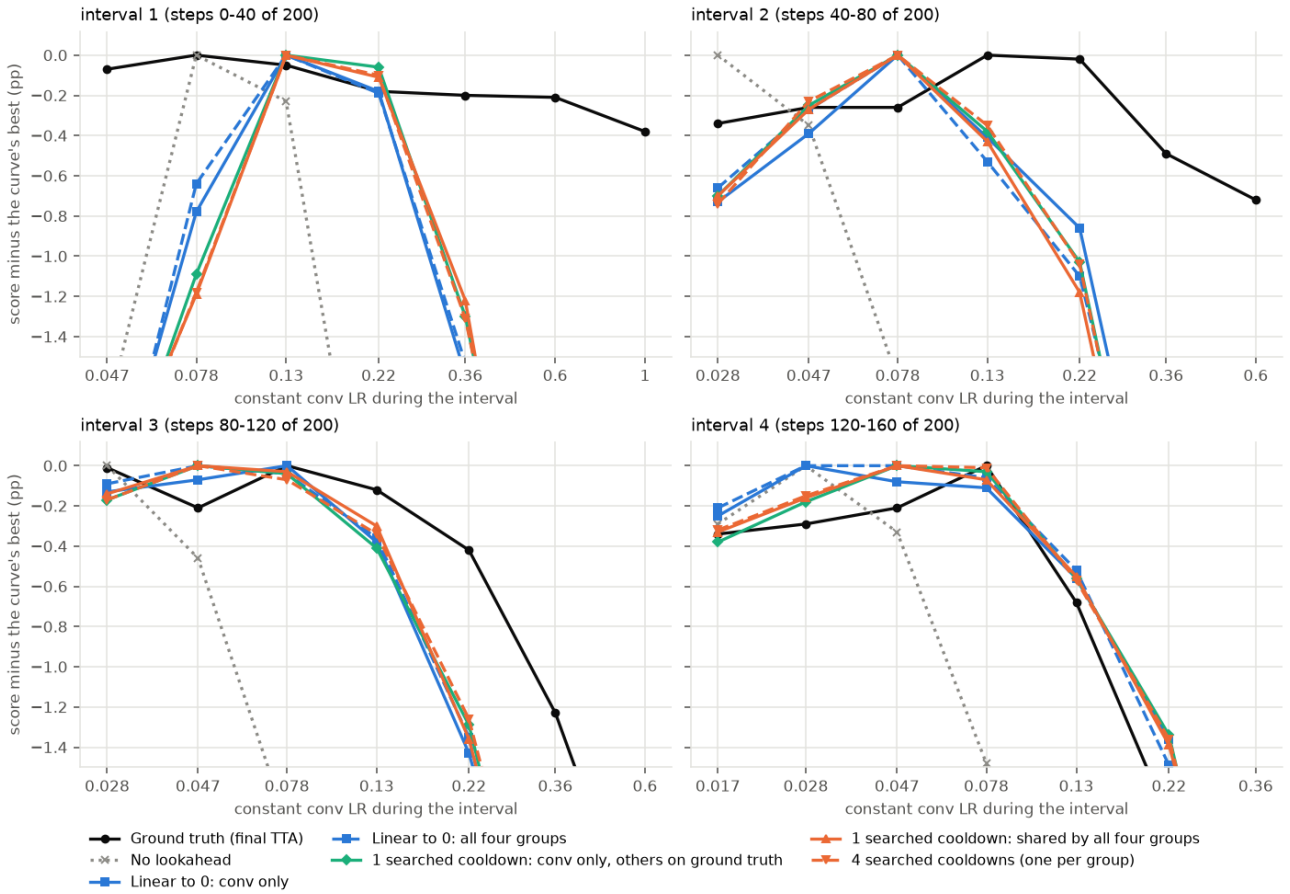


Figure 1. Each method's score per candidate, relative to its own best (the point at 0 is its pick).

Interval 1: score per candidate conv LR (bold: each row's best)

Method	0.047	0.078	0.13	0.22	0.36	0.6	1
Ground truth (final TTA)	93.99	**94.06**	94.01	93.88	93.86	93.85	93.68
No lookahead	81.57	**83.35**	83.12	80.45	68.80	56.07	35.18
Linear to 0: conv only	89.04	90.59	**91.37**	91.19	89.78	87.38	84.39
Linear to 0: all four groups	88.85	90.63	**91.27**	91.08	89.74	87.41	84.10
1 searched cooldown: conv only, others on ground truth	88.58	89.83	**90.92**	90.86	89.62	87.45	85.95
1 searched cooldown: shared by all four groups	88.65	89.70	**90.89**	90.78	89.67	87.16	85.13
4 searched cooldowns (one per group)	88.79	89.89	**91.07**	90.97	89.76	87.65	86.13

Candidate	Conv only: chosen m (seeds)	Shared: chosen m (seeds)	4 searched: chosen (conv, head, norm_bias, whiten_bias) (seeds)
0.047	0.6 x1	0.6 x1	(1, 1, 0.6, 0.36) x1
0.078	0.36 x1	0.36 x1	(0.36, 1, 1, 0.22) x1
0.13	0.22 x1	0.22 x1	(0.22, 1, 0.22, 0.22) x1
0.22	0.22 x1	0.22 x1	(0.22, 0.36, 0.22, 1) x1
0.36	0.13 x1	0.13 x1	(0.13, 0.047, 0.13, 0.13) x1
0.6	0.078 x1	0.078 x1	(0.078, 1, 0.36, 0.078) x1
1	0.078 x1	0.078 x1	(0.078, 0.6, 0.078, 0.6) x1

Interval 2: score per candidate conv LR (bold: each row's best)

Method	0.028	0.047	0.078	0.13	0.22	0.36	0.6
Ground truth (final TTA)	93.76	93.84	93.84	**94.10**	94.08	93.61	93.38
No lookahead	**91.12**	90.77	89.51	87.43	82.45	71.27	57.08
Linear to 0: conv only	92.25	92.59	**92.98**	92.57	92.12	90.18	87.94
Linear to 0: all four groups	92.27	92.66	**92.93**	92.40	91.83	90.11	87.78
1 searched cooldown: conv only, others on ground truth	92.16	92.61	**92.86**	92.48	91.83	89.82	87.93
1 searched cooldown: shared by all four groups	92.18	92.61	**92.88**	92.45	91.70	89.73	87.84
4 searched cooldowns (one per group)	92.20	92.71	**92.94**	92.59	91.90	89.87	87.98

Candidate	Conv only: chosen m (seeds)	Shared: chosen m (seeds)	4 searched: chosen (conv, head, norm_bias, whiten_bias) (seeds)
0.028	0.6 x1	0.6 x1	(0.6, 0.6, 1, 0.6) x1
0.047	0.36 x1	0.36 x1	(0.36, 1, 0.047, 0.36) x1
0.078	0.22 x1	0.22 x1	(0.22, 0.13, 0.22, 0.22) x1
0.13	0.22 x1	0.22 x1	(0.22, 1, 0.36, 0) x1
0.22	0.13 x1	0.13 x1	(0.13, 1, 1, 0.078) x1
0.36	0.13 x1	0.13 x1	(0.13, 0.6, 0, 0.047) x1
0.6	0.078 x1	0.078 x1	(0.078, 0.047, 0.13, 0.078) x1

Interval 3: score per candidate conv LR (bold: each row's best)

Method	0.028	0.047	0.078	0.13	0.22	0.36	0.6
Ground truth (final TTA)	93.99	93.79	**94.00**	93.88	93.58	92.77	91.54
No lookahead	**92.59**	92.13	90.81	87.68	82.32	71.78	56.20
Linear to 0: conv only	93.26	93.32	**93.39**	93.01	91.96	90.45	87.63
Linear to 0: all four groups	93.28	**93.37**	93.33	93.01	92.02	90.23	87.59
1 searched cooldown: conv only, others on ground truth	93.16	**93.33**	93.29	92.92	92.04	90.31	88.61
1 searched cooldown: shared by all four groups	93.13	**93.27**	93.24	92.97	91.91	90.29	88.64
4 searched cooldowns (one per group)	93.18	**93.35**	93.28	93.01	92.09	90.44	88.76

Candidate	Conv only: chosen m (seeds)	Shared: chosen m (seeds)	4 searched: chosen (conv, head, norm_bias, whiten_bias) (seeds)
0.028	0.36 x1	0.22 x1	(0.22, 0.13, 1, 0.22) x1
0.047	0.22 x1	0.36 x1	(0.22, 1, 1, 0.36) x1
0.078	0.22 x1	0.22 x1	(0.22, 1, 0.36, 0.22) x1
0.13	0.22 x1	0.22 x1	(0.22, 1, 0.22, 0.22) x1
0.22	0.22 x1	0.13 x1	(0.22, 0.36, 1, 0.13) x1
0.36	0.13 x1	0.13 x1	(0.13, 0.078, 0.6, 0.13) x1
0.6	0.078 x1	0.13 x1	(0.078, 0.36, 0.13, 0.13) x1

Interval 4: score per candidate conv LR (bold: each row's best)

Method	0.017	0.028	0.047	0.078	0.13	0.22	0.36
Ground truth (final TTA)	93.65	93.70	93.78	**93.99**	93.31	92.23	89.58
No lookahead	92.89	**93.18**	92.85	91.70	88.84	83.30	72.23
Linear to 0: conv only	93.55	**93.80**	93.72	93.69	93.24	92.44	90.63
Linear to 0: all four groups	93.55	**93.76**	**93.76**	93.70	93.24	92.27	90.57
1 searched cooldown: conv only, others on ground truth	93.51	93.71	**93.89**	93.86	93.33	92.55	90.70
1 searched cooldown: shared by all four groups	93.56	93.73	**93.89**	93.82	93.34	92.50	90.89
4 searched cooldowns (one per group)	93.62	93.79	**93.94**	93.93	93.37	92.58	90.93

Candidate	Conv only: chosen m (seeds)	Shared: chosen m (seeds)	4 searched: chosen (conv, head, norm_bias, whiten_bias) (seeds)
0.017	0.6 x1	0.6 x1	(0.6, 1, 1, 0.6) x1
0.028	0.6 x1	0.6 x1	(0.6, 1, 0.36, 0.6) x1
0.047	0.36 x1	0.36 x1	(0.36, 0.22, 1, 0.36) x1
0.078	0.22 x1	0.22 x1	(0.22, 0.047, 0.22, 0.22) x1
0.13	0.13 x1	0.13 x1	(0.13, 0.22, 0.13, 0.13) x1
0.22	0.13 x1	0.13 x1	(0.13, 1, 0.078, 0.13) x1
0.36	0.13 x1	0.13 x1	(0.13, 0.078, 1, 0.13) x1

Observations

One seed: single-run noise is about 0.13 pp, so ground-truth differences of that size are not resolved. Interval 1's ground truth is flat (93.85-94.06 from 0.047 to 0.6) and interval 3's is non-monotone (0.028: 93.99, 0.047: 93.79, 0.078: 94.00), so regrets of about 0.2 pp or less at those intervals are within noise.

All five 40-step lookahead variants pick the same LR at intervals 1 and 2, and the three searched-constant variants (conv only, shared, four per group) pick the same LR at every interval. How the other three groups cool down (continuing their ground-truth schedules, one shared multiplier, or their own multipliers) never changed the pick. Four searched multipliers score slightly higher in absolute terms than one but rank the candidates the same way.

The 40-step lookahead picks a smaller LR than the ground truth where the ground truth is clearly peaked: interval 2 (all variants pick 0.078 where 0.13 is best, 0.26 pp regret) and interval 4 (0.028-0.047 where 0.078 is best, 0.21-0.29 pp). No lookahead is worse at intervals 2 and 4 (0.34 / 0.29 pp) and always picks the smallest or second-smallest candidate after interval 1.

Linear decay and searched constants differ only at intervals 3 and 4, in opposite directions (linear conv-only is better at interval 3, searched constants at interval 4), both by about 0.2-0.3 pp: not resolved with one seed.

Experiment 2. Hparam importance from scratch

Question

How important is each of the four LRs when they are searched jointly from scratch, and does the answer depend on the main interval length?

Setup

Item	Value
Model / data	baseline_bs2000 except the four trained LRs (conv, head, norm_bias, whiten_bias), which are constant for the first L training steps from initialization
Main interval L	1, 2, 5, 10, 40 steps; the score is taken right after the L steps (no lookahead)
Joint search	From scratch: every LR starts at 1.0. Scan each LR over x10^k, k = -3..3 (others fixed), keep the best, snap to the 0.6^k grid, then coordinate-climb on the grid (one step up / down, move while the score improves) until no LR moves: the center
Sweep	From the center, each LR alone over 5 grid points on each side (others at the center). Offset +1 = one grid step larger (x1/0.6), -1 = one step smaller (x0.6)
Seed / score	1 seed (0); test-set TTA accuracy (%); the test cross-entropy loss is logged too

Joint-search centers

L (steps)	Center: conv, head, norm_bias, whiten_bias	Center TTA	Evaluations
1	0.13, 60, 0.00078, 60	25.43	73
2	0.22, 1, 1.7, 1300	36.89	71
5	0.078, 0.006, 2100, 1300	52.00	76
10	0.078, 0.017, 2100, 13	59.47	70
40	0.078, 2100, 280, 2100	84.60	73

Importance: TTA drop from the center (pp) at 1 step / 2 steps (mean of both sides) / worst within 5 steps

LR	L = 1	L = 2	L = 5	L = 10	L = 40
conv	0.58 / 2.01 / 16.42	1.00 / 2.90 / 19.69	3.20 / 7.96 / 32.96	2.02 / 4.11 / 29.37	1.95 / 5.26 / 49.82
head	0.03 / 0.14 / 0.29	0.02 / 0.02 / 0.08	0.10 / 0.12 / 0.18	0.32 / 0.49 / 0.67	0.54 / 1.65 / 62.56
norm_bias	0.00 / 0.00 / 0.00	0.05 / 0.02 / 0.08	0.51 / 1.46 / 28.09	1.12 / 2.77 / 28.84	0.43 / 0.84 / 6.98
whiten_bias	0.01 / 0.03 / 0.09	0.15 / 0.85 / 4.48	0.56 / 1.02 / 3.45	0.51 / 0.58 / 2.00	0.46 / 0.96 / 2.44

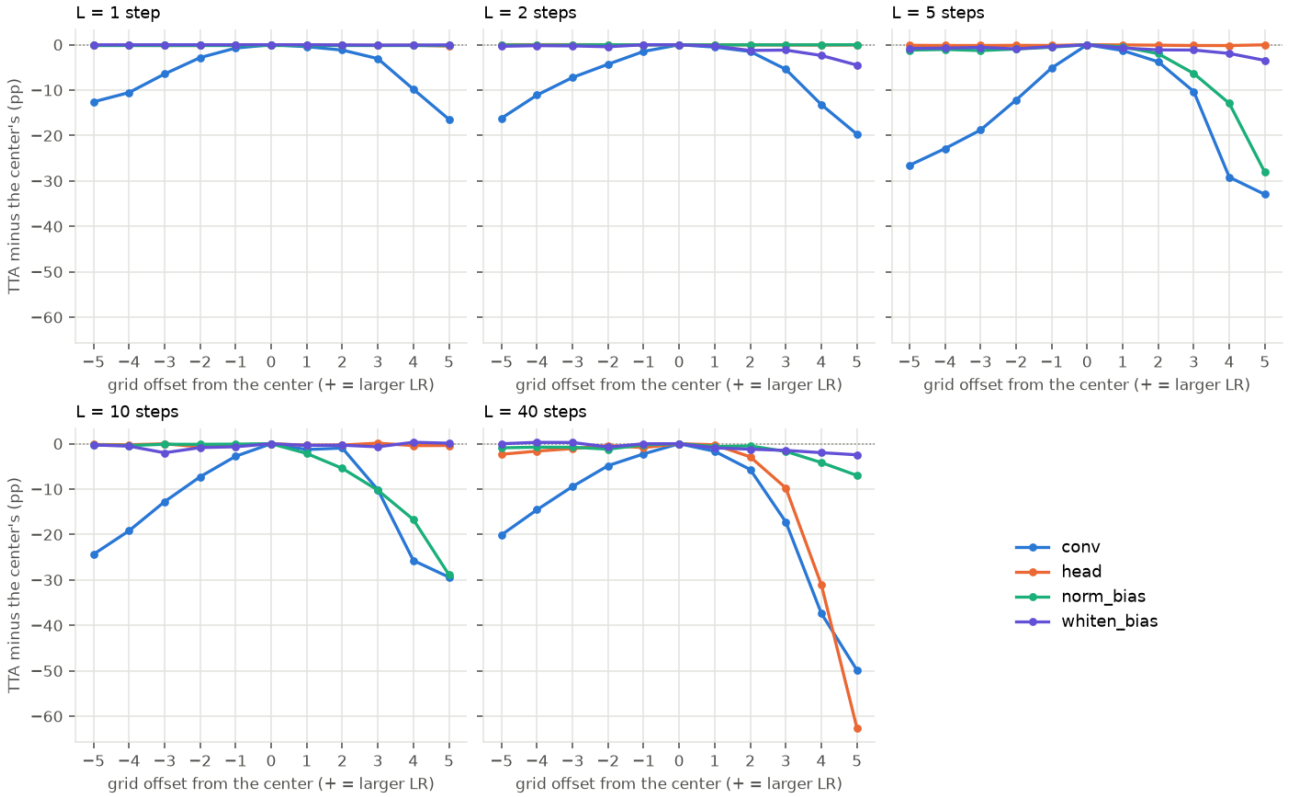


Figure 2. One-at-a-time sweeps around each L's joint-search center.

L = 1: TTA minus the center's (pp) at each grid offset (center LR in brackets; center TTA 25.43)

LR	-5	-4	-3	-2	-1	+0	+1	+2	+3	+4	+5
conv (0.13)	-12.58	-10.53	-6.44	-2.84	-0.73	+0.00	-0.43	-1.18	-3.06	-9.80	-16.42
head (60)	-0.15	-0.14	-0.17	-0.16	-0.03	+0.00	-0.03	-0.11	-0.16	-0.12	-0.29
norm_bias (0.00078)	+0.00	+0.00	+0.00	+0.00	+0.00	+0.00	+0.00	+0.00	+0.00	+0.00	+0.00
whiten_bias (60)	-0.03	-0.02	-0.01	-0.01	-0.01	+0.00	-0.01	-0.05	-0.09	-0.08	-0.04

L = 2: TTA minus the center's (pp) at each grid offset (center LR in brackets; center TTA 36.89)

LR	-5	-4	-3	-2	-1	+0	+1	+2	+3	+4	+5
conv (0.22)	-16.19	-11.09	-7.21	-4.29	-1.48	+0.00	-0.52	-1.50	-5.41	-13.19	-19.69
head (1)	+0.01	+0.01	+0.02	+0.00	-0.01	+0.00	-0.03	-0.03	-0.05	-0.08	-0.04
norm_bias (1.7)	-0.07	-0.08	-0.05	-0.01	-0.04	+0.00	-0.05	-0.03	-0.03	-0.02	-0.01
whiten_bias (1300)	-0.34	-0.20	-0.26	-0.44	-0.01	+0.00	-0.29	-1.26	-1.18	-2.37	-4.48

L = 5: TTA minus the center's (pp) at each grid offset (center LR in brackets; center TTA 52.00)

LR	-5	-4	-3	-2	-1	+0	+1	+2	+3	+4	+5
conv (0.078)	-26.55	-22.87	-18.78	-12.17	-5.13	+0.00	-1.28	-3.75	-10.31	-29.17	-32.96
head (0.006)	-0.13	-0.17	-0.17	-0.14	-0.16	+0.00	-0.04	-0.10	-0.17	-0.18	-0.02
norm_bias (2100)	-1.19	-1.04	-1.29	-0.94	-0.54	+0.00	-0.47	-1.97	-6.33	-12.92	-28.09
whiten_bias (1300)	-0.81	-0.75	-0.58	-0.91	-0.41	+0.00	-0.71	-1.13	-1.17	-1.92	-3.45

L = 10: TTA minus the center's (pp) at each grid offset (center LR in brackets; center TTA 59.47)

LR	-5	-4	-3	-2	-1	+0	+1	+2	+3	+4	+5
conv (0.078)	-24.35	-19.17	-12.77	-7.28	-2.76	+0.00	-1.28	-0.95	-10.13	-25.72	-29.37
head (0.017)	-0.17	-0.23	-0.02	-0.67	-0.31	+0.00	-0.34	-0.30	+0.13	-0.42	-0.38
norm_bias (2100)	-0.29	-0.39	-0.12	-0.13	-0.09	+0.00	-2.15	-5.42	-10.23	-16.70	-28.84
whiten_bias (13)	-0.26	-0.53	-2.00	-0.85	-0.67	+0.00	-0.36	-0.32	-0.68	+0.32	+0.11

L = 40: TTA minus the center's (pp) at each grid offset (center LR in brackets; center TTA 84.60)

LR	-5	-4	-3	-2	-1	+0	+1	+2	+3	+4	+5
conv (0.078)	-20.08	-14.54	-9.42	-4.83	-2.24	+0.00	-1.67	-5.69	-17.22	-37.36	-49.82
head (2100)	-2.33	-1.63	-1.07	-0.40	-0.85	+0.00	-0.23	-2.90	-9.74	-31.04	-62.56
norm_bias (280)	-0.90	-0.75	-0.77	-1.20	-0.24	+0.00	-0.62	-0.48	-1.69	-4.15	-6.98
whiten_bias (2100)	+0.00	+0.30	+0.27	-0.72	-0.01	+0.00	-0.91	-1.21	-1.47	-1.97	-2.44

Experiment 3. Representation quality of schedule families

Question

How close to the baseline can each family of conv LR schedules get, under no transform and under the ramp, reverse ramp and wave transforms, when searched by the same global search?

Setup

Item	Value
Model / data	baseline_bs2000; only the conv LR schedule is searched, the other groups stay on the baseline. The transform multiplies the conv LR the search specifies
Conditions	No transform; ramp x 4^(2x-1) (x0.25 -> x4); reverse ramp x 4^(1-2x) (x4 -> x0.25); wave x 2^sin(2 pi x) (x1 -> x2 -> x0.5 -> x1); x = fraction of training
Families	5 constant intervals (5 values); linear decay to 0 (initial LR); power decay lr x (1-t)^p (lr, p; p = 1 is linear); 2 knots at t = 0, 0.5; 3 knots at t = 0, 1/3, 2/3; 5 knots at t = 0, 0.2, 0.4, 0.6, 0.8 (knot values; linear between knots, 0 at the end)
Search	Generalized global_neighbour (global_family): coordinate search over the family's parameters on multiplicative grids (LR values x0.6, p x0.8), one grid step at a time, moving while the score improves; every candidate is a full training run on seed 0 from the same initial state
Start	The oracle spec (baseline conv schedule divided by the transform) projected onto the family: interval means or knot values snapped to the grid; linear / power start at 0.22 with p = 1
Evaluation	Start and best schedules retrained on held-out seeds 100-107 (mean test TTA; not part of the search)

References (held-out TTA, seeds 100-107)

	No transform	Ramp	Reverse ramp	Wave
Baseline linear decay, no transform (the target)	93.92	93.92	93.92	93.92
Baseline spec kept under the transform (do-nothing)	93.92	93.42	93.57	93.72

Held-out TTA of the best schedule (start = the projected oracle before search; search cost)

Family	No transform	Ramp	Reverse ramp	Wave
5 constant intervals	**93.62** (start 93.52; 18 runs, 182 s)	**93.65** (start 93.32; 28 runs, 285 s)	**93.86** (start 93.81; 28 runs, 279 s)	**93.82** (start 93.11; 22 runs, 222 s)
Linear decay (1 param)	**93.86** (start 93.92; 3 runs, 31 s)	**93.42** (start 93.42; 3 runs, 31 s)	**93.59** (start 93.57; 4 runs, 41 s)	**93.91** (start 93.72; 3 runs, 31 s)
Power decay (lr, p)	**93.86** (start 93.92; 8 runs, 81 s)	**93.95** (start 93.42; 14 runs, 141 s)	**93.59** (start 93.57; 6 runs, 61 s)	**93.91** (start 93.72; 5 runs, 51 s)
2 knots (t = 0, 0.5)	**93.86** (start 93.86; 5 runs, 51 s)	**93.77** (start 93.14; 9 runs, 91 s)	**93.88** (start 93.85; 8 runs, 83 s)	**93.83** (start 93.72; 12 runs, 121 s)
3 knots (t = 0, 1/3, 2/3)	**93.86** (start 93.94; 12 runs, 121 s)	**93.68** (start 93.74; 10 runs, 101 s)	**93.78** (start 93.90; 13 runs, 131 s)	**94.06** (start 93.90; 7 runs, 73 s)
5 knots (t = 0, 0.2, ..., 0.8)	**93.94** (start 93.97; 13 runs, 131 s)	**93.96** (start 93.91; 17 runs, 170 s)	**93.91** (start 93.91; 11 runs, 111 s)	**93.94** (start 93.97; 11 runs, 111 s)

Best parameter values (LR values in order; power: lr, p)

Family	No transform	Ramp	Reverse ramp	Wave
5 constant intervals	0.13, 0.22, 0.078, 0.047, 0.028	0.36, 0.13, 0.078, 0.028, 0.0036	0.028, 0.078, 0.13, 0.078, 0.047	0.078, 0.078, 0.13, 0.078, 0.017
Linear decay (1 param)	0.13	0.22	0.078	0.13
Power decay (lr, p)	0.13, 0.64	0.36, 2	0.078, 1	0.13, 1
2 knots (t = 0, 0.5)	0.22, 0.13	0.6, 0.078	0.078, 0.078	0.13, 0.078
3 knots (t = 0, 1/3, 2/3)	0.13, 0.22, 0.047	0.36, 0.13, 0.047	0.078, 0.078, 0.047	0.13, 0.078, 0.13
5 knots (t = 0, 0.2, ..., 0.8)	0.13, 0.13, 0.13, 0.078, 0.047	1, 0.36, 0.22, 0.047, 0.017	0.047, 0.078, 0.078, 0.13, 0.13	0.13, 0.078, 0.078, 0.13, 0.078

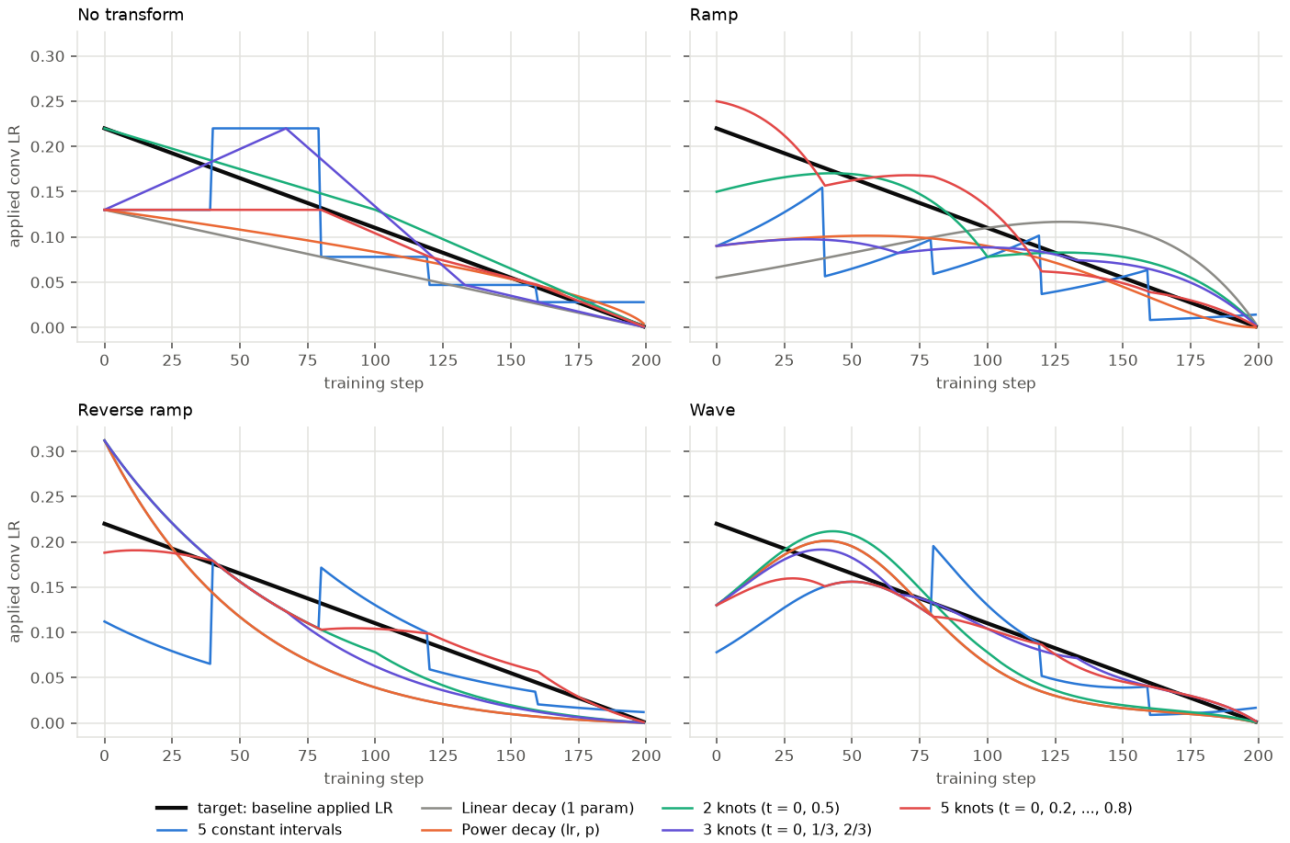


Figure 3. Applied conv LR (spec x transform) of each family's best schedule; the black line is the baseline's applied LR, which a perfect search would reproduce.

Observations

Held-out means over 8 seeds have a standard error of about 0.045 pp, so differences below about 0.1 pp are not resolved. Target: the baseline's 93.92 on the same seeds.

5 knots is the only family at the target in all four conditions (93.91-93.96), and its start, the oracle projected onto the family, is already there (93.91-93.97): 5 knots on the grid represent all four target schedules.

5 constant intervals cannot represent them. The projected oracle scores 93.52 / 93.32 / 93.81 / 93.11 (no transform / ramp / reverse ramp / wave); search raises that to 93.62 / 93.65 / 93.86 / 93.82, still 0.1-0.3 pp short without a transform and under ramp. Its last value stays at 0.028 without a transform: a constant final interval at that level does not anneal.

Linear decay matches the target without a transform (93.86-93.92) and under wave (93.91), but no initial LR compensates a ramp (93.42) or a reverse ramp (93.59): the shape is wrong, not the scale. Power decay fixes the ramp (93.95 with p = 2, lr 0.36: decaying faster) but not the reverse ramp (93.59): (1-t)^p cannot rise early, which the reverse ramp needs.

2 and 3 knots land in between (93.68-94.06), with no consistent ordering across conditions.

Single-seed search sometimes leaves a good start for a worse held-out schedule (no transform: linear 93.92 -> 93.86, 3 knots 93.94 -> 93.86, 5 constant intervals moved its first value from 0.22 to 0.13): the search optimizes seed 0's test score, the same winner's curse as before.